



Design and Development of an Arduino-Based Optical Computed Tomography (CT) Scanning System

PHYS 437A - Research Project | Life Physics | Joan Gabrielle Basa

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Problem Statement



Problem

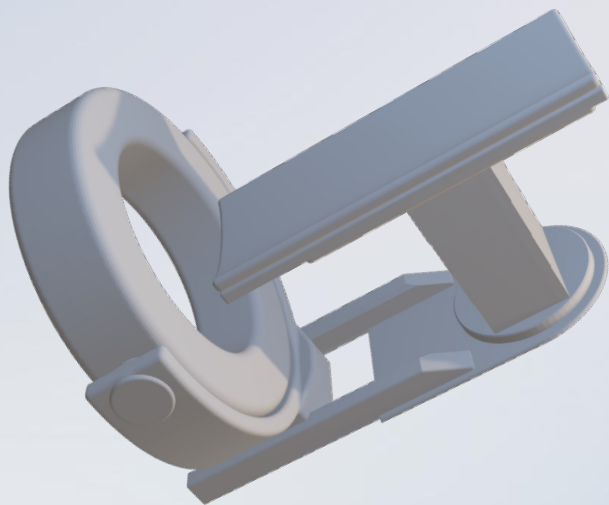
Inaccessibility of traditional CT scanners

Motivation

Dynamic 3D imaging and materials analysis

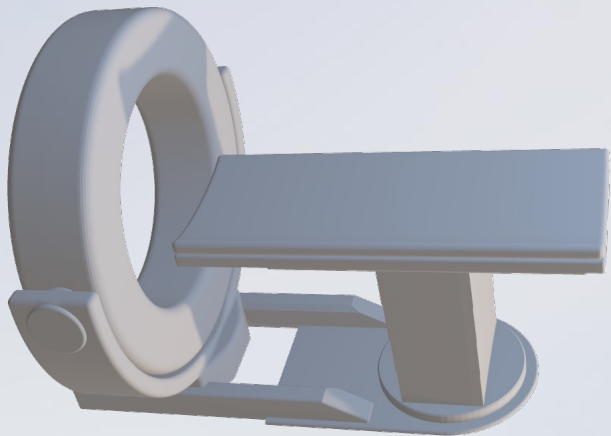
Goal

To build a compact visible-light CT system using optical projections and tomographic algorithms





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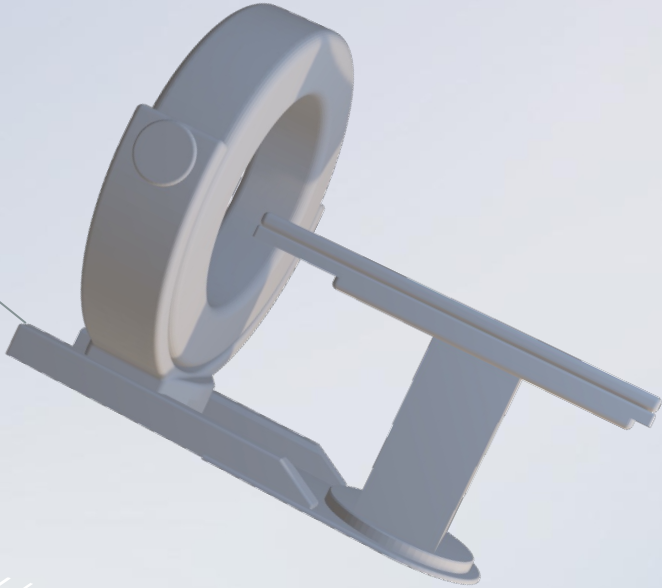
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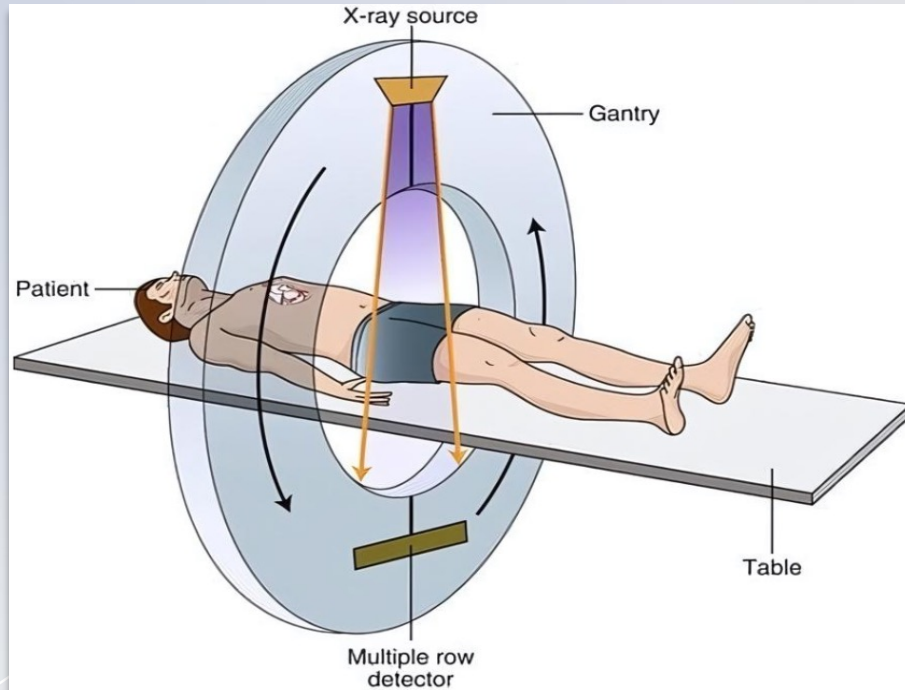
Dynamic 3D imaging and materials analysis

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CT Instrumentation^[24]



01

Source (X-Ray Tube)

Emits a focused, high-energy X-ray beam

02

Detectors

Captures X-rays converts them into signals

03

Gantry

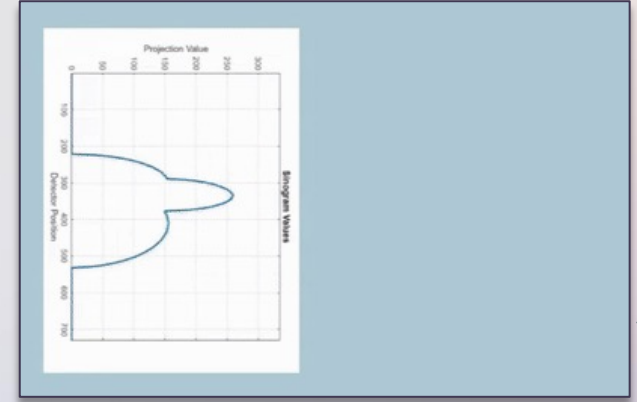
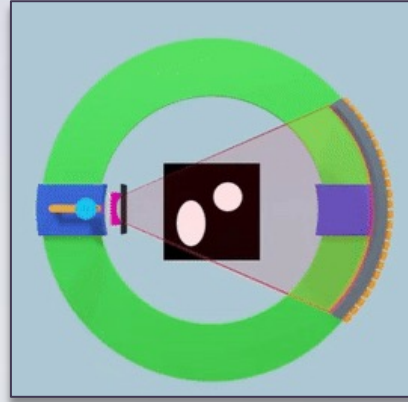
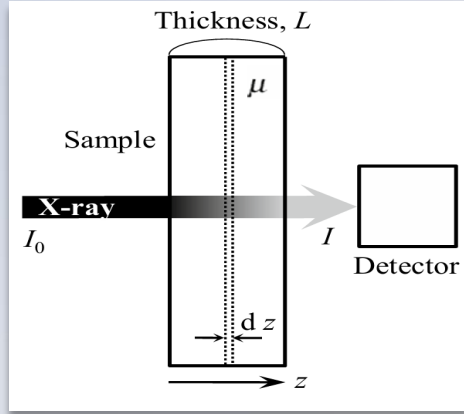
Holds the x-ray tube and detectors

04

Table (or Stage)

Moves through the gantry for axial or helical scanning

CT Image Formation and Reconstruction^[24,26]



$$I_d = I_0 \exp \left[- \int_0^d \mu(s; \bar{E}) ds \right] \rightarrow g_d = -\ln \left(\frac{I_d}{I_0} \right) = - \int_0^d \mu(s; \bar{E}) ds \rightarrow g(l, \theta) = \iint_{-\infty}^{\infty} \mu(x, y) \delta(x \cos \theta + y \sin \theta - l) dx dy$$

Projections

Radon
Transform

Sinograms

Inverse Radon
Transform

Reconstructions



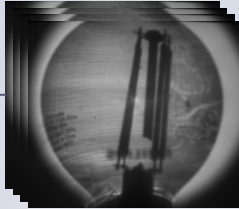
Project Overview

Acquisition

Settings

Reconstruction

Hardware

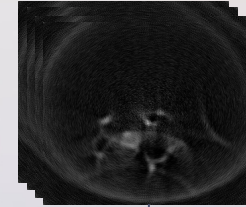


Projections

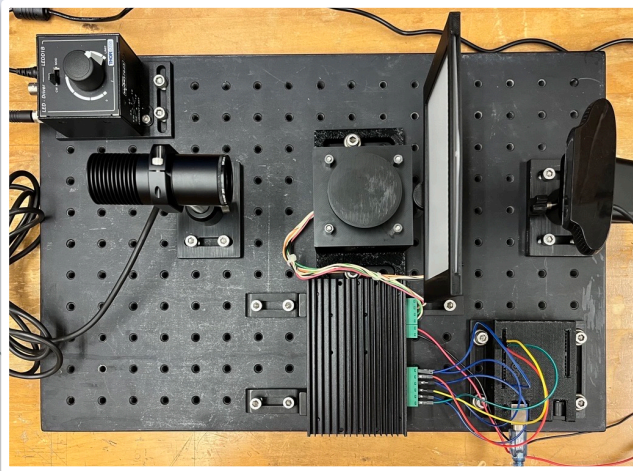
Noise Reduction
Contrast Enhancement
Alignment and Tilt
Adjustments

Sinograms

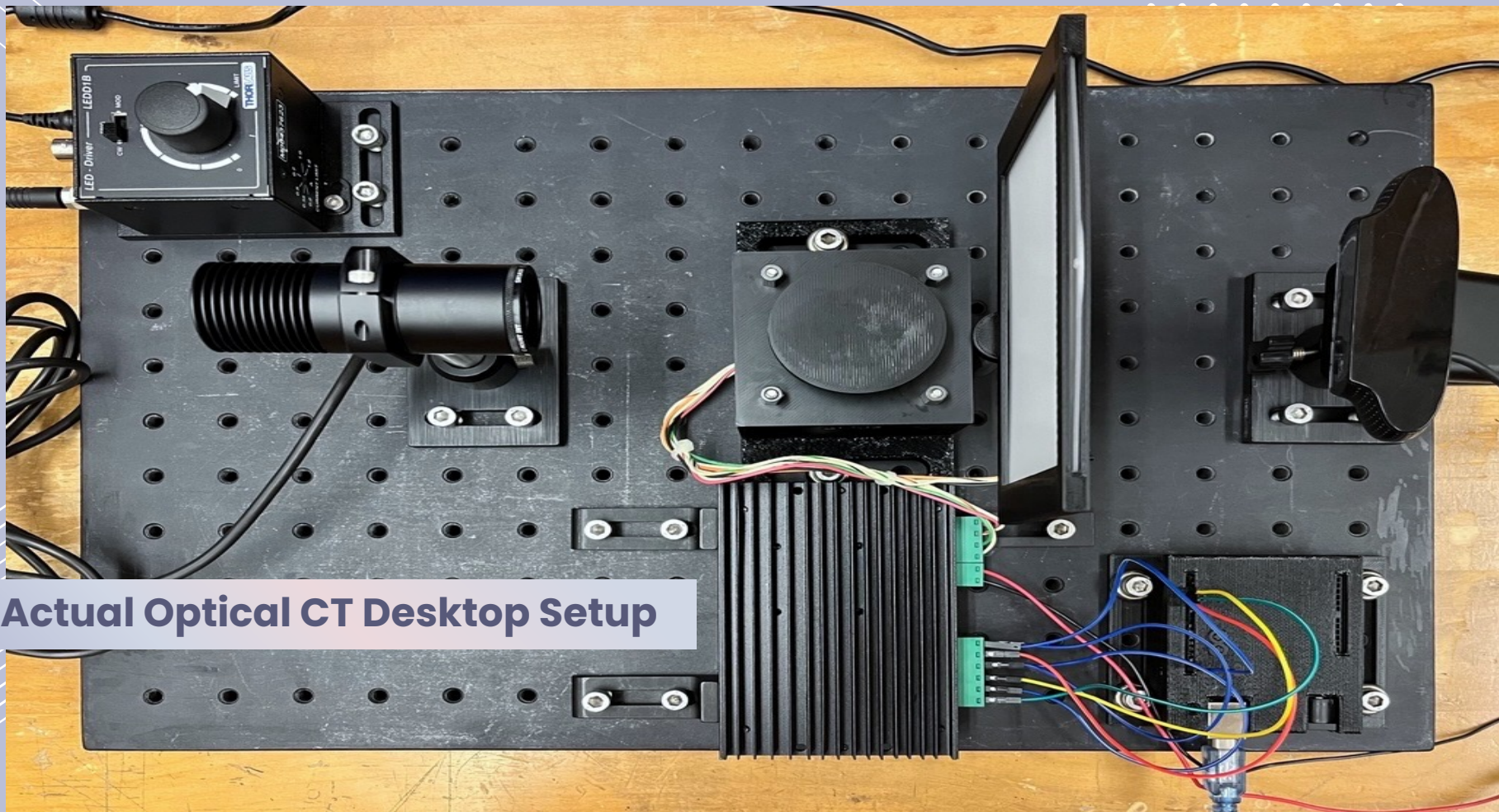
Filtered Backprojection



3D Render



	Medical CT	Optical CT
Source	X-ray tube	Visible light
Sample	Opaque	Transparent or semi-transparent materials
Detector	Scintillating material	Diffusing filter
Rotation	Source + Detector	Sample



Actual Optical CT Desktop Setup

System Setup

Visible Light Source

Rotating Sample Stage

Flat Projection Screen

CCD or Web Camera

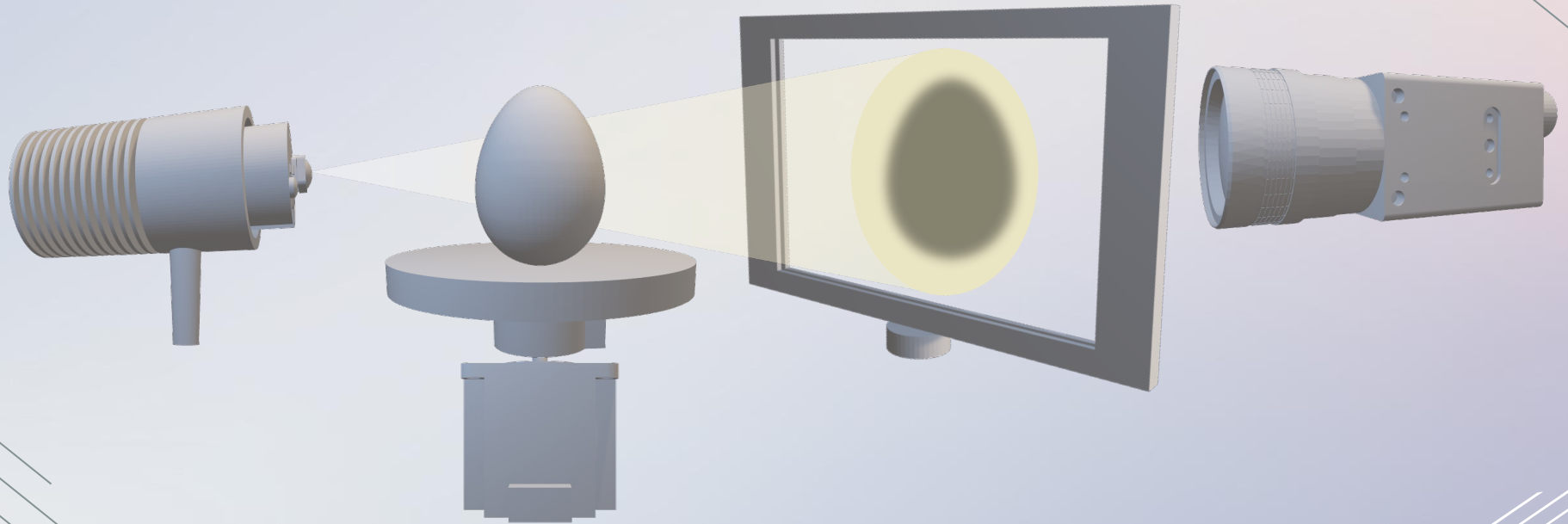


Image Processing and Reconstruction^[17-18,27,38]

Pre-processing

- Reduce noise
- Enhance contrast
- Fix alignment and tilt
- Invert projection stack (optional)

Post-processing

- 3D Slicer or Volume Viewer (ImageJ) for 3D rendering of tomograms*

MATLAB I

Construct Sinograms

MATLAB II

Center Sinograms (ROI)

MATLAB III

Construct Tomograms

MATLAB IV

Resize Tomograms

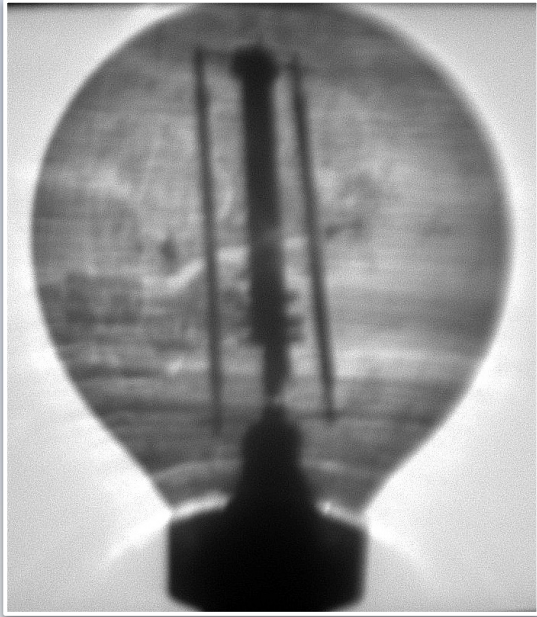
MATLAB V

3D Volume Rendering*

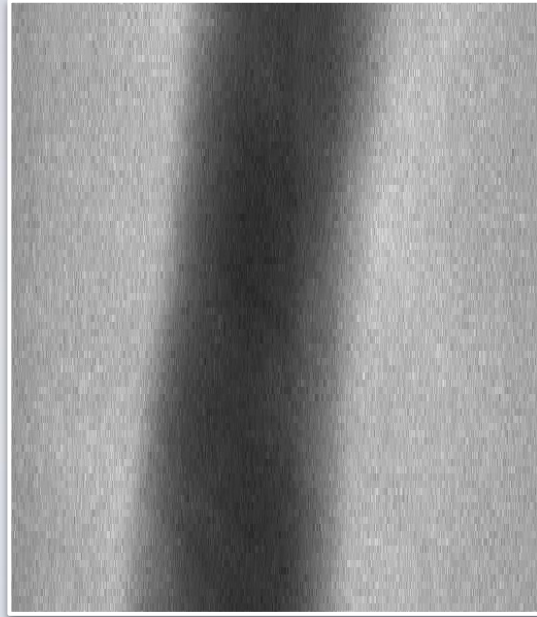
**if and only if there is a huge stack of tomograms to be processed*



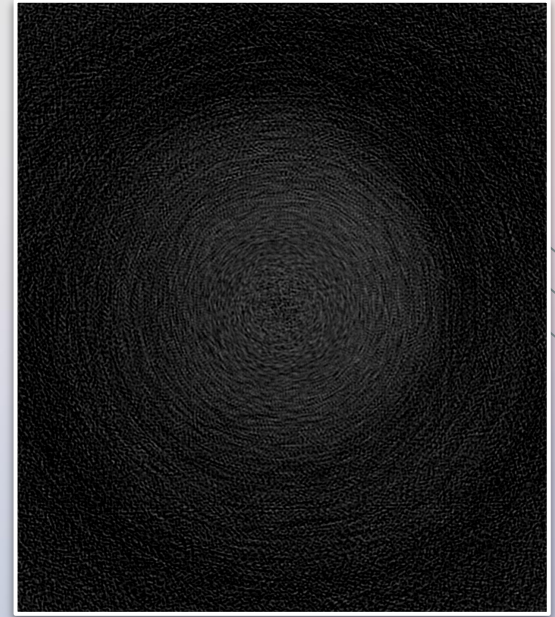
Results: Visualizations



Projections



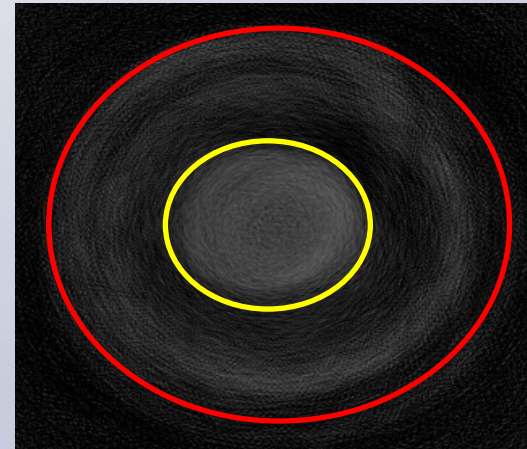
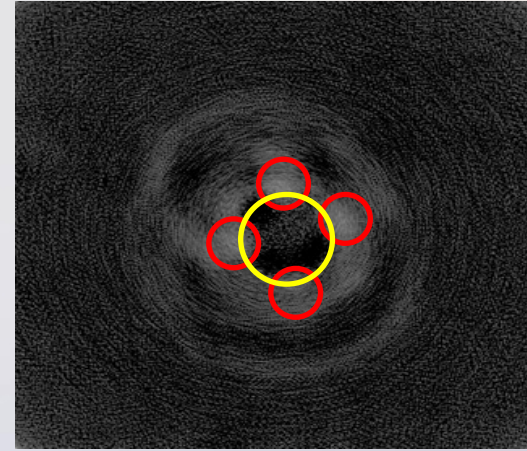
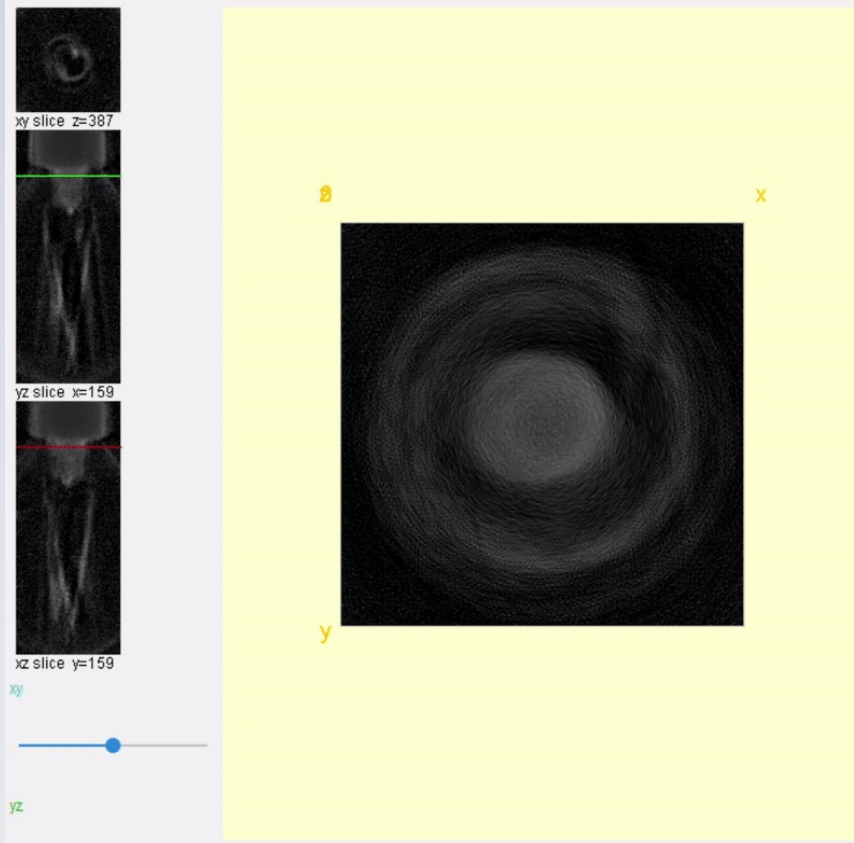
Sinograms



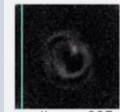
Tomograms



Results: 2D Slice (XY)



Results: 2D Slice (XZ or YZ)



xy slice z=387



yz slice x=159

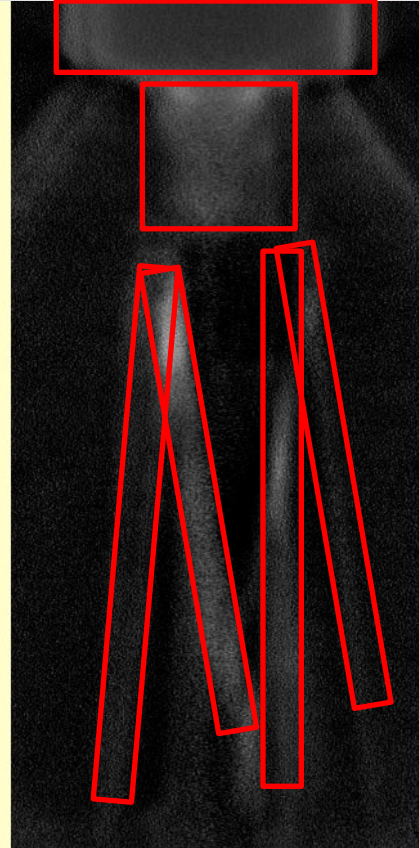


xz slice y=159

xy



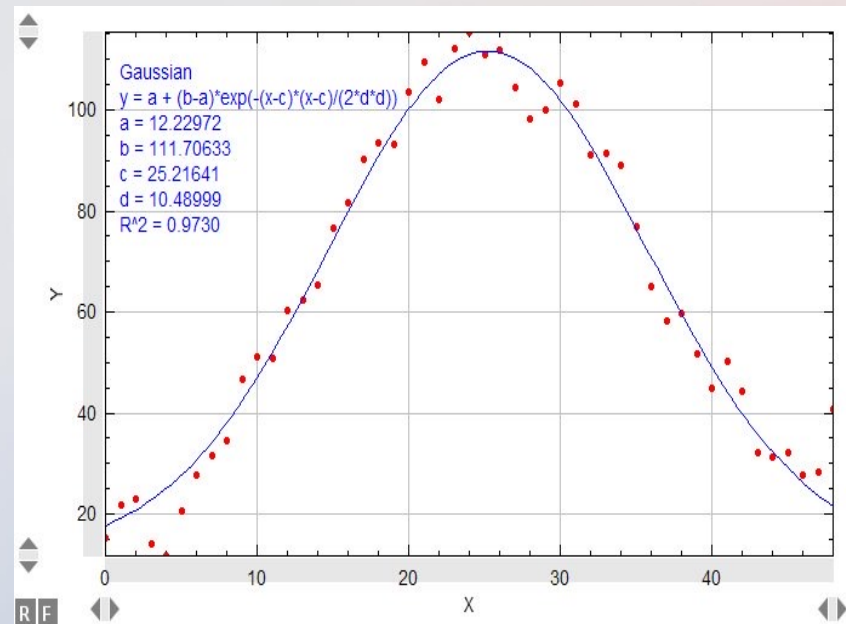
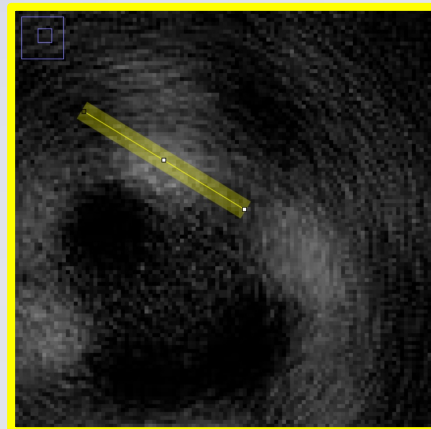
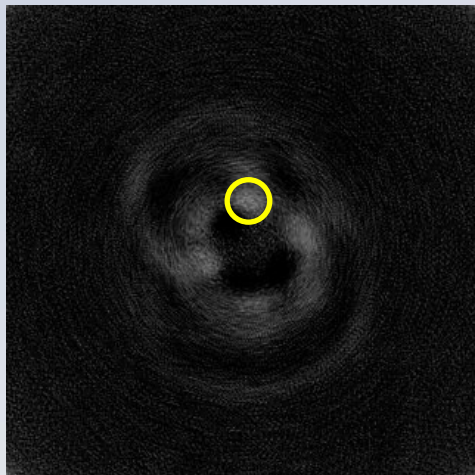
yz



Volume Viewer



Characterization: FWHM^[24,38]



FWHM $\approx 59.38 \mu\text{m}$

Challenges and Limitations

Artifacts

Scattering and
refractions

Depth

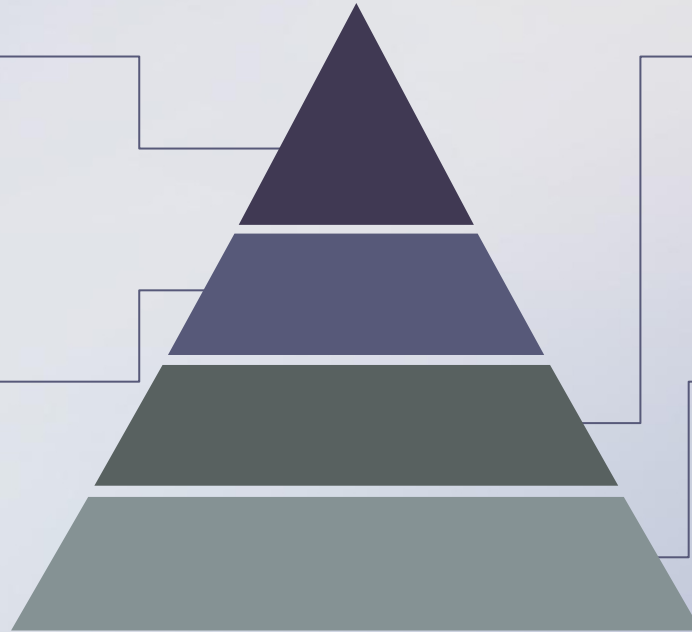
Limited penetration for
more opaque samples

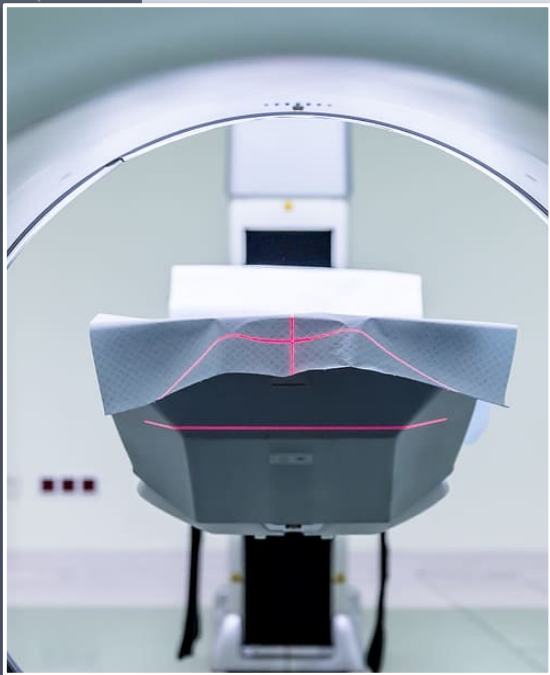
Hardware

Misalignment, vibrations
or wobble, ambient light

Software

Computational limits of
software programs





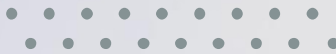
Future Perspectives

Optimizations

Exploring light sources, filters, other reconstruction software for improved process workflow and reconstructions

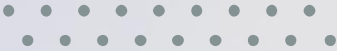
Experiments

- Dynamic imaging of periodic motion
 - Analysis of 3D printed bolus materials under mechanical or thermal stress



References

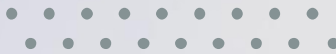
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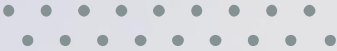




References

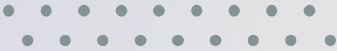
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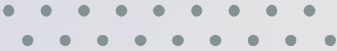
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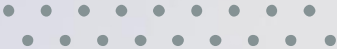




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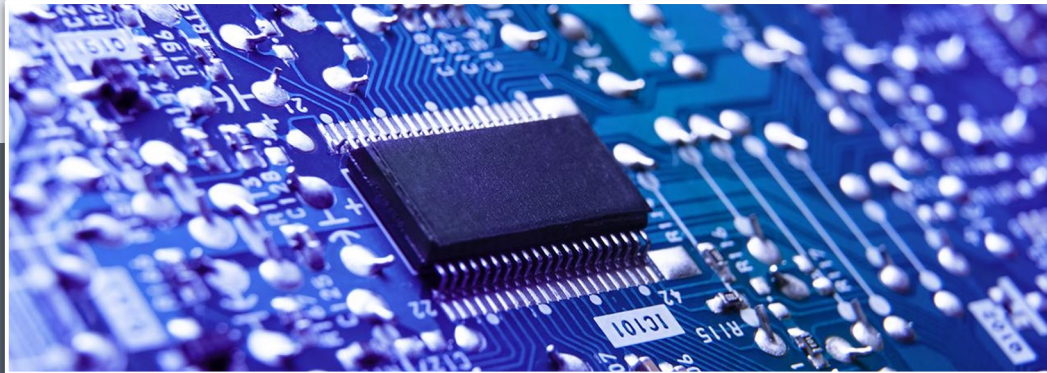
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Thank you

Any questions, concerns, or insights?

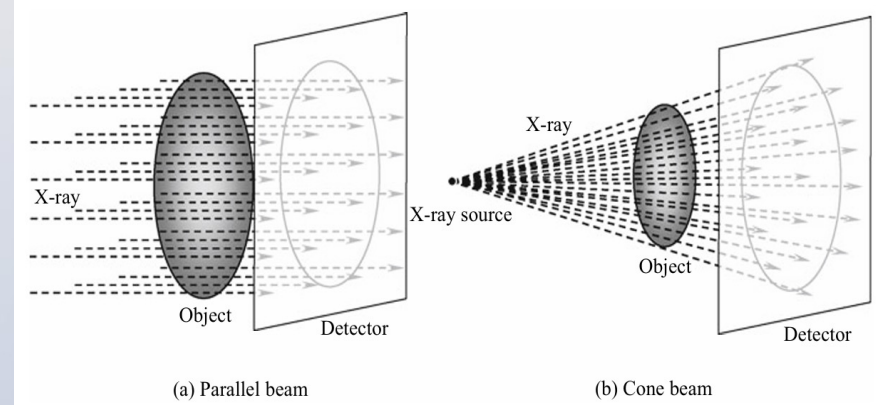
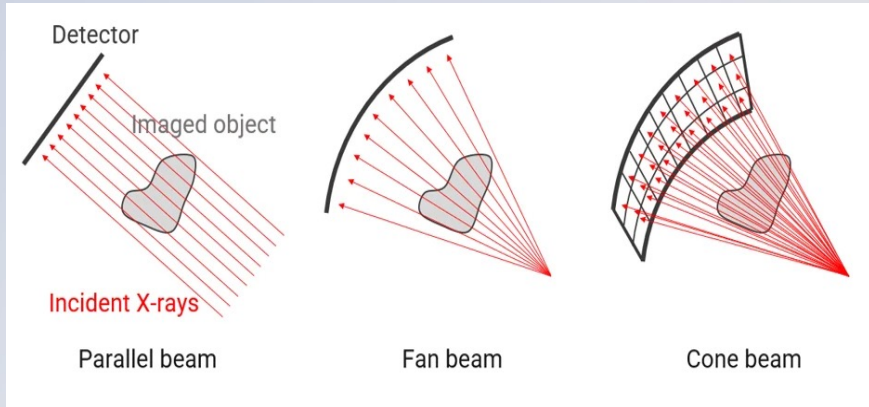


Generations of CT Scanners^[24]

Comparison of CT Generations

Generation	Source	Source Collimation	Detector	Detector Collimation	Source–Detector Movement	Advantages	Disadvantages
1G	Single x-ray tube	Pencil beam	Single	None	Move linearly and rotate in unison	Scattered energy is undetected	Slow
2G	Single x-ray tube	Fan beam, not enough to cover FOV	Multiple	Collimated to source direction	Move linearly and rotate in unison	Faster than 1G	Lower efficiency and larger noise because of the collimation in detectors
3G	Single x-ray tube	Fan beam, enough to cover FOV	Many	Collimated to source direction	Rotate in synchrony	Faster than 2G, continuous rotation using a slip ring	More expensive than 2G, low efficiency
4G	Single x-ray tube	Fan beam covers the FOV	Stationary ring of detectors	Cannot collimate detectors	Detectors are fixed, source rotates	Higher efficiency than 3G	High scattering since detectors are not collimated
5G (EBCT)	Many tungsten anodes in single large tube	Fan beam	Stationary ring of detectors	Cannot collimate detectors	No moving parts	Extremely fast, capable of stop-action imaging of beating heart	High cost, difficult to calibrate
6G (Helical CT)	3G/4G	3G/4G	3G/4G	3G/4G	3G/4G plus linear patient table motion	Fast 3D images	A bit more expensive
7G (Multiple-row detector CT)	Single x-ray tube	Cone beam	Multiple arrays of detectors	Collimated to source direction	3G/4G/6G motion	Fast 3D images	Expensive

Beam Geometry^[32,36]



Lens Tube System^[35]

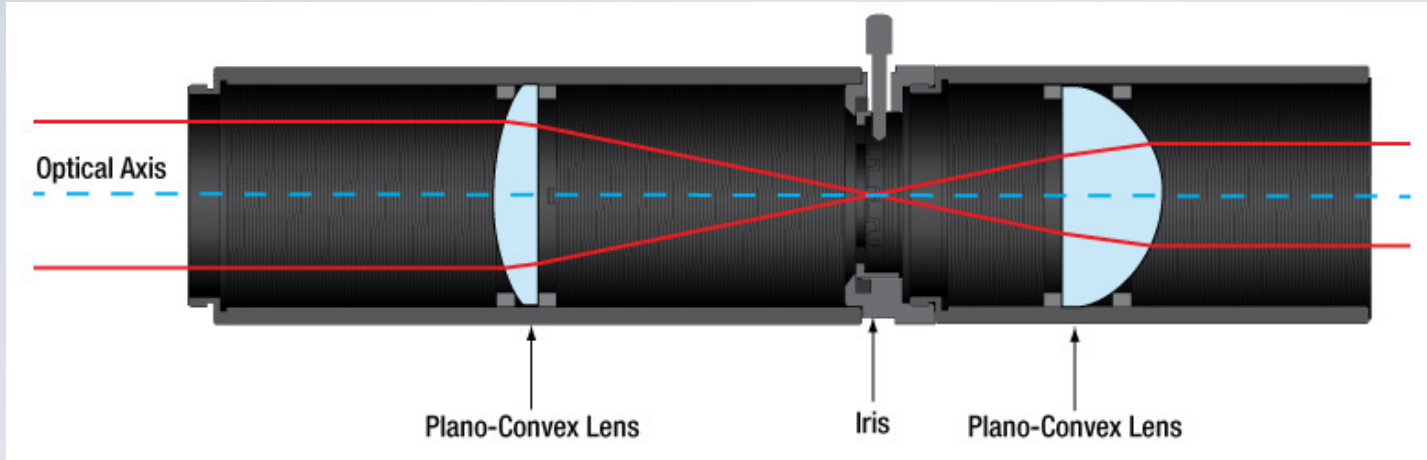


Table G8.1 Specifications

Item #	Info ^a	Correlated Color Temperature ^b	Housing Type ^c	LED Output Power (Min / Typ.) ^{b,d}	Bandwidth (FWHM)	Irradiance (Typ.) ^e	Max Current (CW)	Forward Voltage (Typ.)	Viewing Angle (Full Angle at Half Max)	Recommended Drivers ^f
MWWHL4 ^g		3000 K (Warm White)		570 mW / 640 mW	N/A	9.4 $\mu\text{W}/\text{mm}^2$	1000 mA	3.0 V	120°	LEDD1B, DC40, UPLED, DC2200, DC4100 ^h , or DC4104 ^h

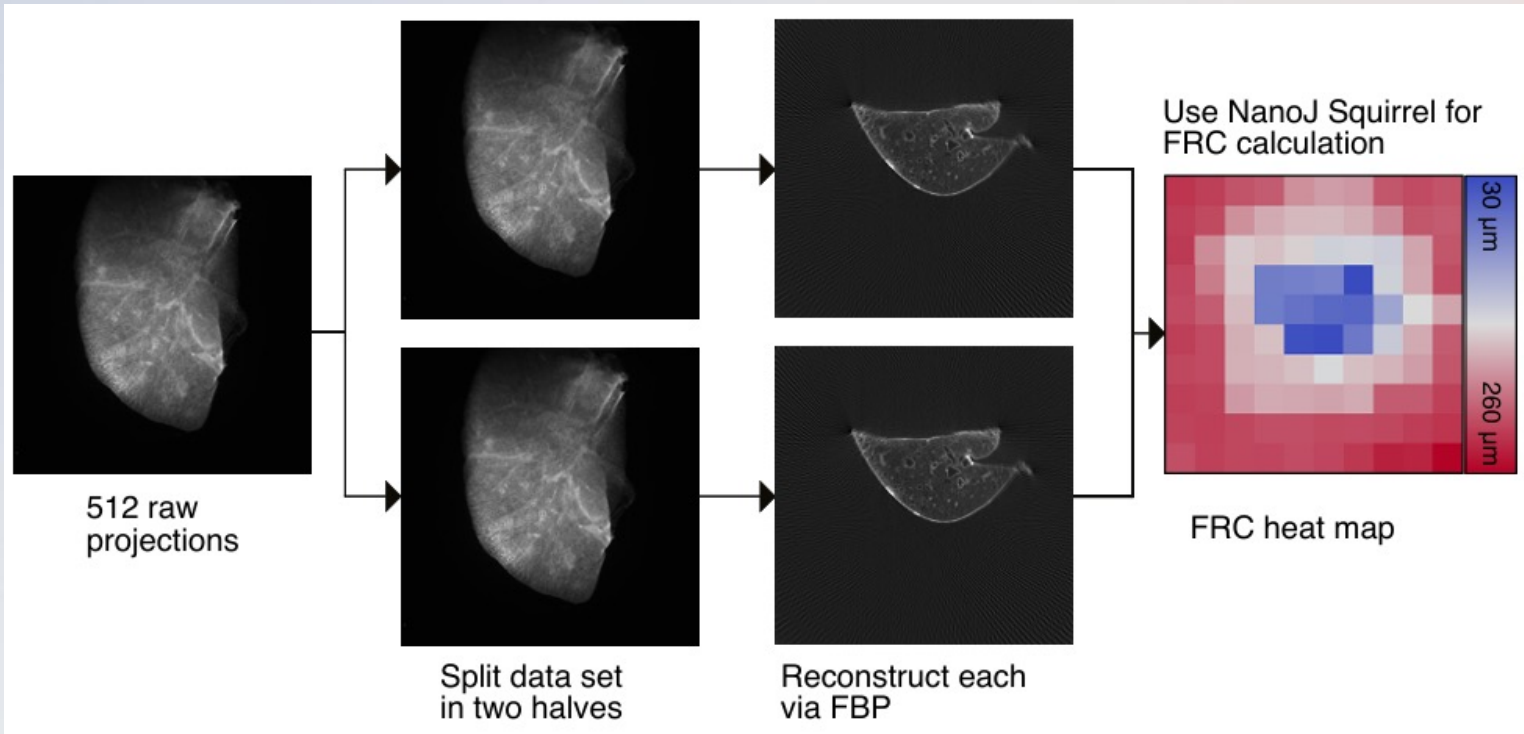


FBP Filter Types^[37]



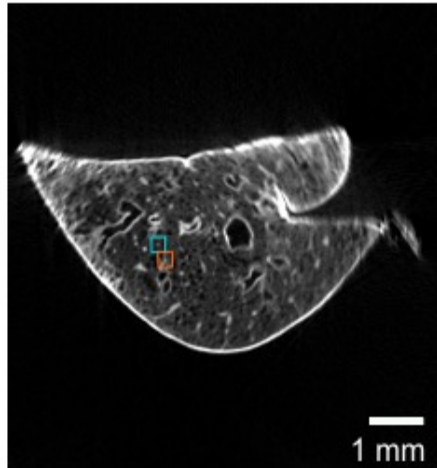
	Description	Use Case(s)	Image
Ramp (or Ram-Lak)	Enhances spatial resolution by boosting high frequencies	• When maximum spatial resolution is needed • Examples: bone or dental imaging	
Hamming	Smooth low-pass filter; excellent for noise suppression, esp. in noisy or low-flux conditions	• When image smoothness is more important • Example: Pediatric CT or low-dose chest scans	
Shepp-Logan	Modifies the Ramp filter by applying a sinc function to reduce high-frequency noise and ringing	• For good spatial resolution with less noise • Example: Brain or soft tissue imaging in low-dose CT	

Fourier Ring Correlation (FRC)^[38]

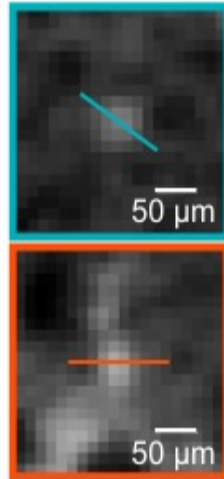


Full-Width at Half Maximum (FWHM)^[38]

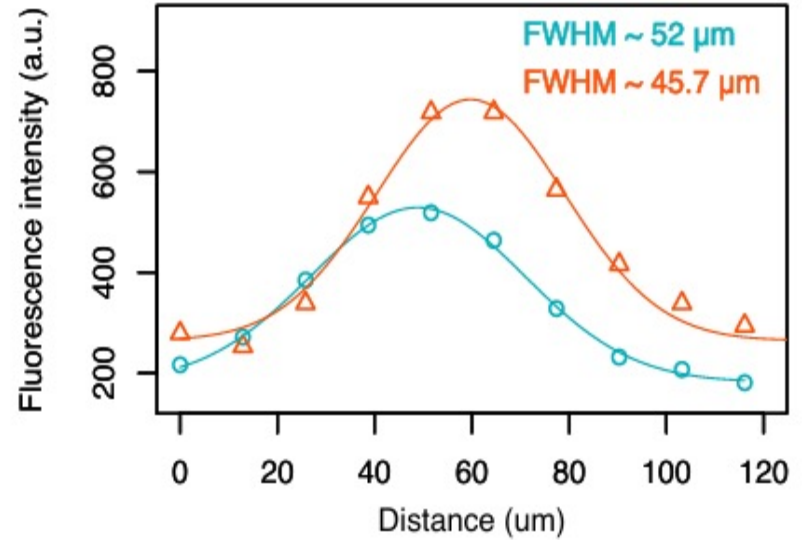
Reconstructed slice



Line profile



Gaussian Fit



MATLAB: Construct Sinograms^[17-18]

```
Sing = zeros(nFrames,dx,3);  
Sing = uint8(Sing);
```

```
offset_y = 1;  
frjmp = 1; %frame jump = every [frjmp] frame  
cnt = 1;  
  
for kk = 1:offset_y:dy
```

```
i = 1; % Counter for image name  
for k=1:frjmp:nFrames  
  
    img=read(obj,k);  
    img_crop = imcrop(img,[xl yt dx dy]);  
    img_crop_sing = imcrop(img_crop,[0 (kk) (dx) 0]); % second crop
```

MATLAB: Construct Tomograms^[17-18]

```
for jj = 1:cxy
%   F = fileNames(jj)
    F = fullfile(fileFolder,fileNames(jj));

    im = imread(F);
    im = imrotate(im,270);
```

```
I1 = iradon(grrS,1);
```



CT Projection-Slice Theorem^[11,24,32,36-37]



Collect projections (360-degree rotation)



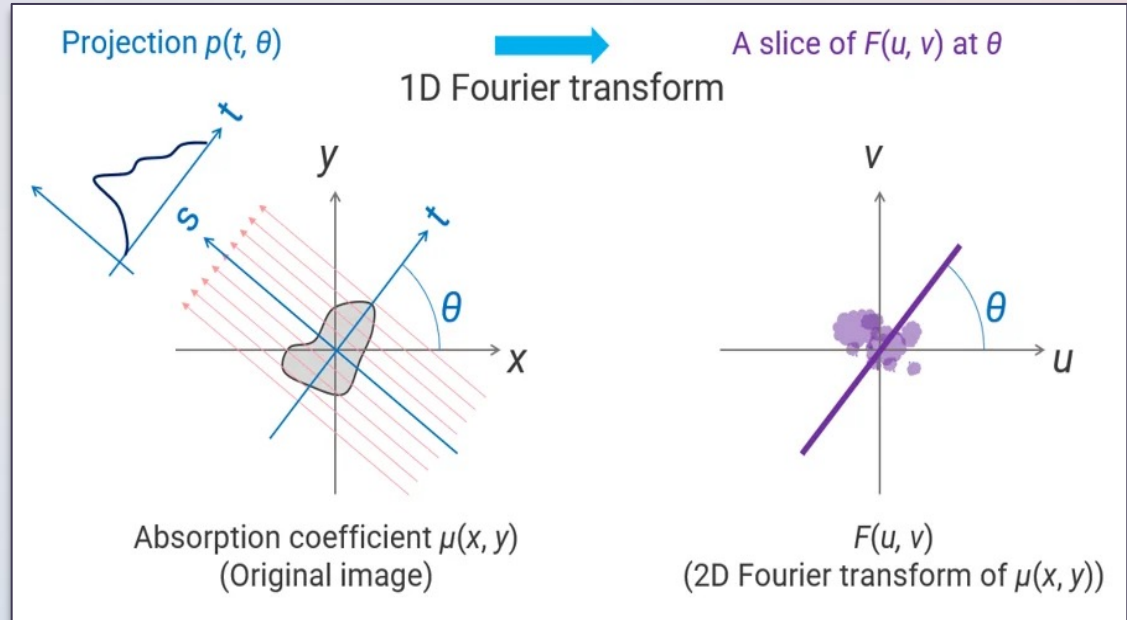
Perform Fourier transform of projections



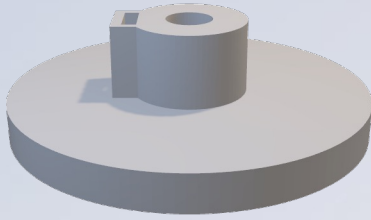
Add all projections in the Fourier space



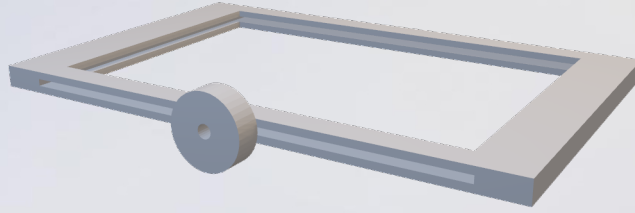
Inverse Fourier transform to obtain reconstruction



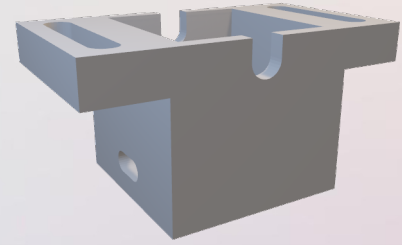
Hardware: 3D Models and Prints^[7,23]



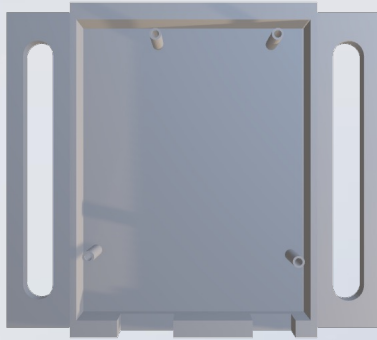
Stage Base



Screen Holder



Motor Mount



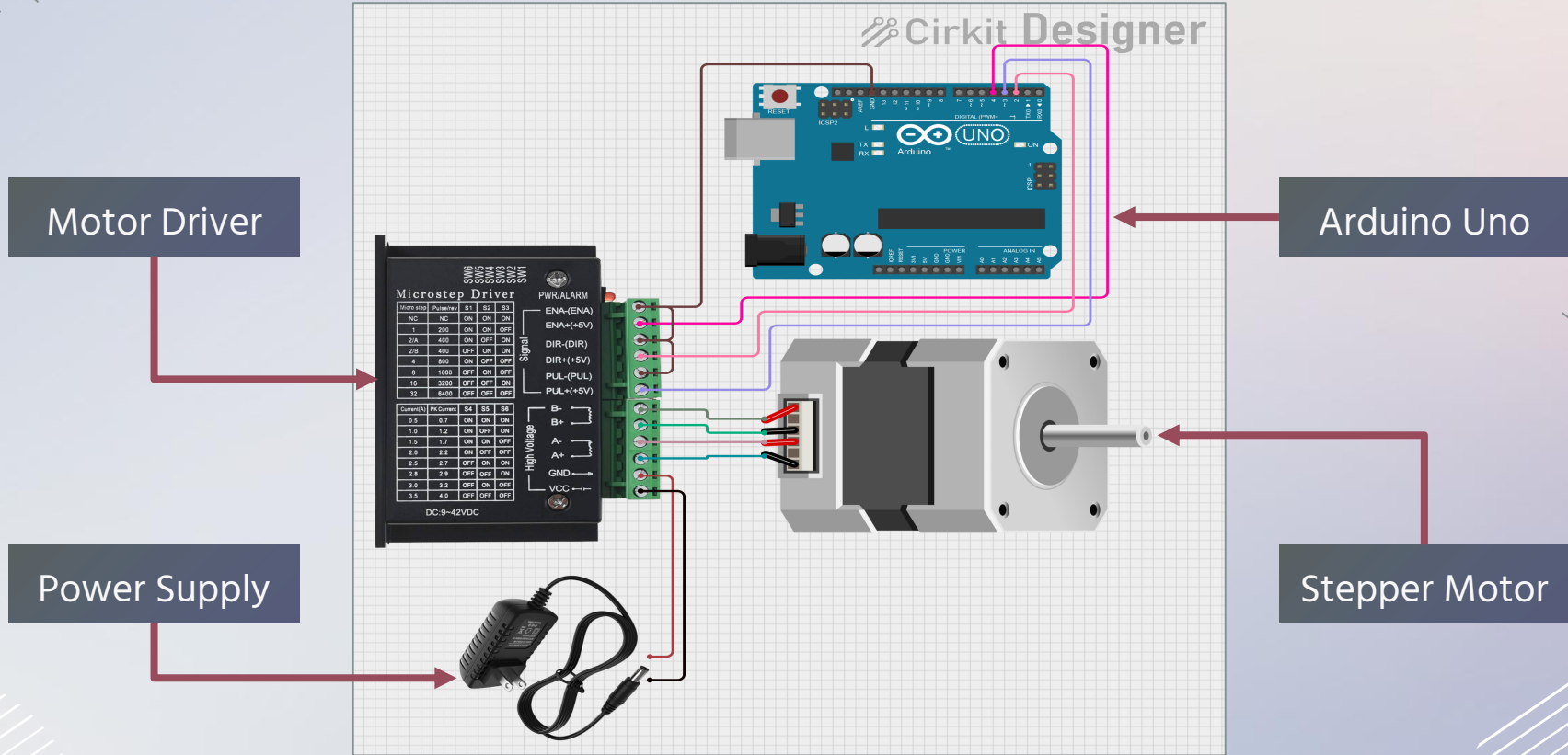
Arduino Mount



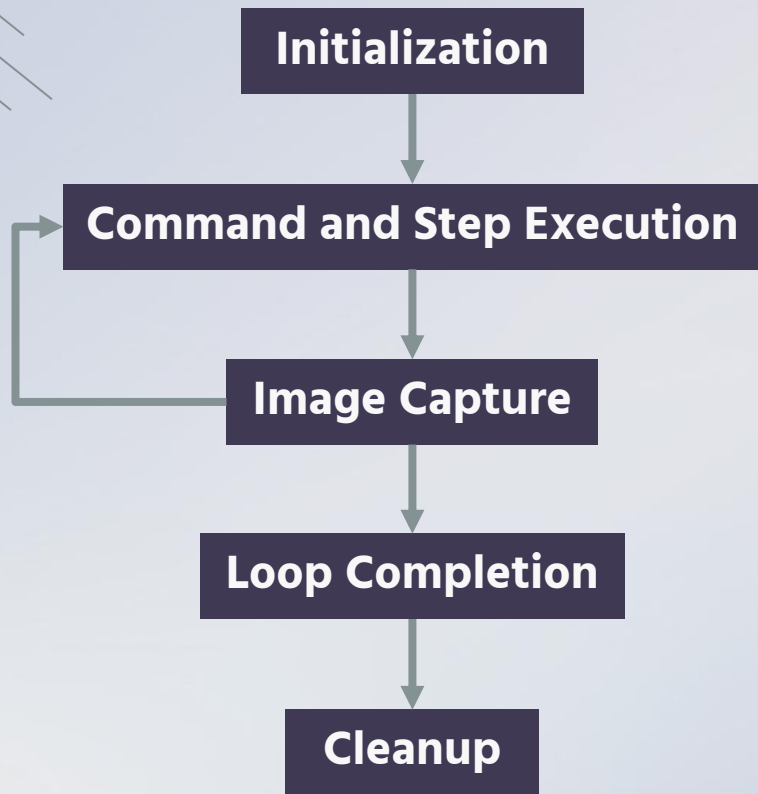
CCD Mount



Hardware: Wiring Diagram



Stepper Motor Control + Image Capture^[1,12]



```
const int dirPin = 2;
const int stepPin = 3;
const int enPin = 4;

void setup() {
  pinMode(dirPin, OUTPUT);
  pinMode(stepPin, OUTPUT);
  pinMode(enPin, OUTPUT);
  digitalWrite(enPin, LOW);
  Serial.begin(9600);
}

void loop() {
  if (Serial.available() > 0) {
    char command = Serial.read();

    if (command == 'F') {
      digitalWrite(dirPin, HIGH);
    }
    else if (command == 'R') {
      digitalWrite(dirPin, LOW);
    }
    else if (command == 'S') {
      digitalWrite(stepPin, HIGH);
      delayMicroseconds(500);
      digitalWrite(stepPin, LOW);
      delayMicroseconds(500);
      Serial.println("STEP_DONE");
    }
  }
}
```

C++

```
import cv2
import serial
import time

# --- CONFIG ---
serial_port = 'COM4'
baud_rate = 9600
num_steps = 400
output_folder = 'captured_images'

# --- SETUP ---
arduino = serial.Serial(serial_port, baud_rate, timeout=1)
time.sleep(2)
cam = cv2.VideoCapture(0)

if not cam.isOpened():
    raise Exception("Could not open webcam")

# --- EXECUTE STEPS ---
for step in range(num_steps):
    arduino.write(b'S')
    line = ""

    while "STEP_DONE" not in line:
        line = arduino.readline().decode('utf-8').strip()
        if line:
            print(f"Arduino: {line}")

    ret, frame = cam.read()
    if ret:
        filename = f"{output_folder}/image_{step:03}.jpg"
        cv2.imwrite(filename, frame)
        print(f"Saved {filename}")
    else:
        print(f"Failed to capture image at step {step}")

cam.release()
arduino.close()
print("Done!")
```

Python

CT Filtered Backprojection (FBP)^[11,19,26]

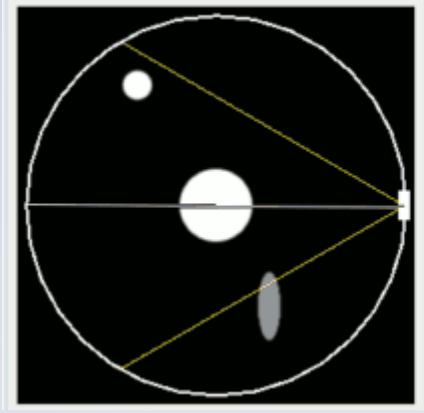


Image Space

Represents the true spatial distribution of physical parameters



Backprojection

"Smearing" projection values back into the image space



FBP

Undoes the blurring caused by simple backprojection



CT Image Formation[11,24,32,36-37]

Attenuation

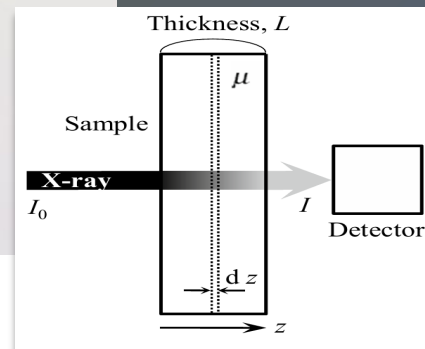
- Materials selectively attenuate beam intensities to form an image
- Line attenuation coefficient (μ)**

Line Integrals

- Total attenuation along a specific path between the source and detector
- CT scanners measure **Radon transform**

CT Numbers (HU or h)

- Normalizes μ values to water ($\mu = 0$)



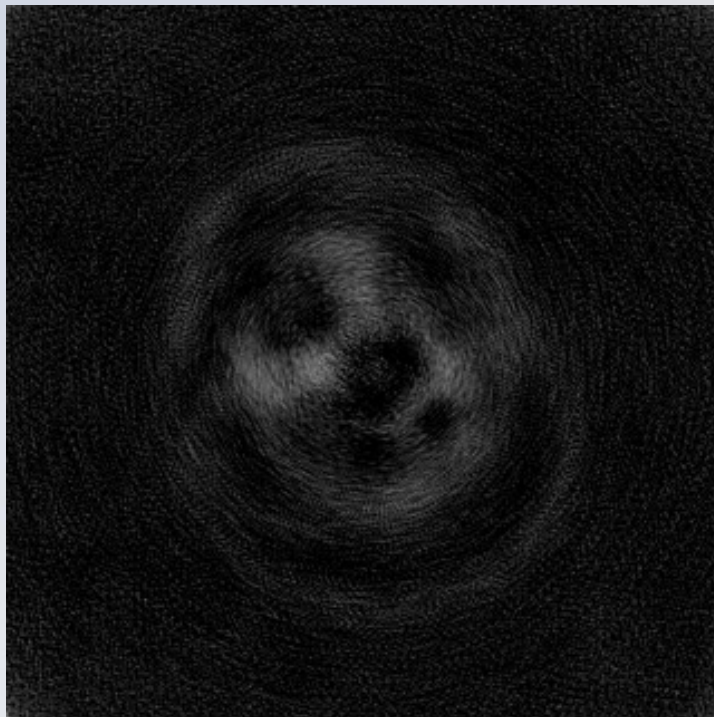
$$I = I_0 e^{-\mu L}$$

$$I_d = I_0 \exp \left[- \int_0^d \mu(s; \vec{E}) ds \right]$$

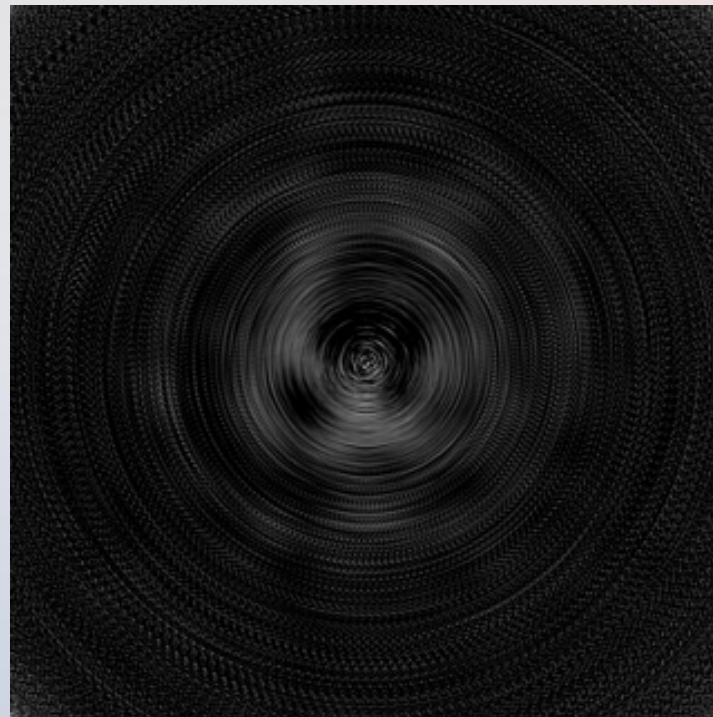
$$g_d = -\ln \left(\frac{I_d}{I_0} \right) = - \int_0^d \mu(s; \vec{E}) ds$$

$$h = 1,000 \times \frac{\mu - \mu_{\text{water}}}{\mu_{\text{water}}}$$

200 vs. 25 Projections



200 Images



25 Images

